INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
Mid Semester Examination 2025-26

Date of Examination: 18/09/25 Session(FN/AN) AN Duration 2 hrs, Marks = 50
Sub No: CS60075 Sub Name: Natural Language Processing
Department/Centre/School : Computer Science and Engineering
Specific charts, graph paper, log book etc. required: NO
Special Instructions (if any) ANSWER ALL questions. No Clarifications! In case of reasonable doubt, make assumptions and state them upfront. Marks will be deducted for sketchy answers and claims without proper reasoning. All parts of a single question should be done at the same place. Keep probability values in fractional forms.

1. Here is a small, 3-sentence training corpus. Assume that the vocabulary only consists of words occurring in it.

`mares eat oats and goats eat oats and little lambs eat ivy .`
`a kid will bleat and eat ivy too .`
`piglets sleep and eat .`

Thus the vocabulary is {`.`, `a`, `and`, `bleat`, `eat`, `goats`, `ivy`, `kid`, `lambs`, `little`, `mares`, `oats`, `piglets`, `sleep`, `too`, `will`}.

(a) Using the maximum likelihood estimation and *unigram* modeling, show the computation for P(`eat oats and eat ivy`).

(b) Using the maximum likelihood estimation and *bigram* modeling, show the computation for P(`eat oats and eat ivy`). When computing the sentence-initial bigram, use the unigram probability.

(c) Now using add-k smoothing with $k = 1$ and *bigram* modeling, show the computation of P(`oats` | `eat`).

[3 + 4 + 2 = 9]

2. Each of the following questions presents a pseudocode implementation of (a part of) a neural network. Looking at the pseudocode, note if there is anything missing and/or incorrect in the given implementation; if yes, then indicate and correct it, and if not, then just record your answer as nothing being wrong. **Mathematically justify** your answers in each case. In case you find the answer incorrect, suggest a **simple fix without changing the functional forms**. Notation: $x(i) \in \mathbb{R}^n$ denotes the i-th training sample (of total m samples), $y(i) \in \mathbb{R}$ denotes the prediction for that i-th training sample, and θ_j-s are the parameters (weights) of the algorithm.

(a) **function** $f(x \in \mathbb{R})$:
  $z \leftarrow \exp(x + 1)$
  $z \leftarrow \log(z)$
  output z

function forward$_{nn}$$(x \in \mathbb{R}^{m \times n})$:
  $\theta \in \mathbb{R}^n \leftarrow \vec{0}$
  $y \in \mathbb{R}^m$
  for $i = 1$ to m:
    $h_{out} \leftarrow \theta^T x^{(i)}$
    $y^{(i)} \leftarrow f(h_{out})$
  output y

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(b) **function** $f(x \in \mathbb{R})$:
$z \leftarrow 2 + \exp(-x)$
$z \leftarrow \frac{2}{z}$
output z

function $\texttt{forward}_{\texttt{nn}}(x \in \mathbb{R}^{m \times n})$:
$\theta \in \mathbb{R}^n \leftarrow \vec{0}$
$y \in \mathbb{R}^m$
for $i = 1$ to m:
$h_{out} \leftarrow \theta^T x^{(i)}$
$y^{(i)} \leftarrow f(h_{out})$
output y

(c) **function** $f(x \in \mathbb{R})$:
$z \leftarrow \sqrt{x}$
output z

function $\texttt{forward}_{\texttt{nn}}(x \in \mathbb{R}^{m \times n})$:
$\theta \in \mathbb{R}^n \leftarrow \vec{0}$
$y \in \mathbb{R}^m$
for $i = 1$ to m:
$h_{out} \leftarrow \theta^T x^{(i)}$
$y^{(i)} \leftarrow f(h_{out})$
output y

[3 + 3 + 4 = 10]

3. By using Viterbi Algorithm, find whether the word `''jumps''` is VB or NN through computing their probabilities for the sentence, `''the cat jumps''`. The transition probabilities and the observation likelihood for this corpus are as follows:

	VB	NN	DT
<S>	0.2	0.6	0.2
VB	0.1	0.3	0.6
NN	0.5	0.3	0.2
DT	0.4	0.4	0.2

	the	cat	jumps
VB	0.05	0.02	0.05
NN	0.05	0.10	0.05
DT	0.3	0.05	0.05

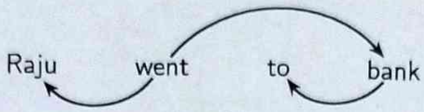
[8]

4. Given the grammar and lexicon below, derive the parse tree using the top-down parsing method for the sentence.
`''The beautiful peacock running on the road''`
N → peacock, road
V → running
Adj → beautiful
Det → The | the
Prep → on
S → NP VP

NP → Pronoun | Noun | Det Adj Noun | NP PP
PP → Prep NP
V → Verb | Aux Verb
VP → V | V NP | V NP NP | V NP PP | VP PP

[3]

5. List all the intermediate configurations and the transitions taken to obtain the parse for the sentence ``Raju went to bank'' using Arc-Eager parsing with the help of the following dependency graph.



[5]

6. Recall that a recurrent neural network (RNN) takes in an input vector x_t and a state vector h_{t-1} and returns a new state vector h_t and an output vector y_t; $h_t = f(w_1 x_t + w_2 h_{t-1} + b_2)$ and $y_t = g(w_3 h_t + b_3)$, where f and g are activations functions. Suppose that f is a sigmoid unit and that the input is always 0. **Draw a plot** (x-range $= [0, 1]$, y-range $= [0, 1]$) showing how the next state h_{t+1} changes with the current state h_t when $w_2 = 3$ and $b_2 = -1$. Assume that $h_{t+1} = h_t$ when $h_t = 0.80$. Hence answer the following questions.

(a) $h_{t+1} = \sigma(\underline{\qquad\qquad})$

(b) Would h_{t+1} be greater than or less than h_t if $h_t = 0.5$? Justify.

(c) Would h_{t+1} be greater than or less than h_t if $h_t = 0.9$? Justify.

(d) What would h_{t+1} look like as we take increasingly longer sequences, i.e., $t \to \infty$?

[2 + 1.5 × 4 = 8]

7. Suppose you have word embeddings for three words: 1, 2, and 3 as follows. Suppose we are using a skip-gram model with softmax probabilities.

$v_1 = (0, 0)$
$v_2 = (1, 1)$
$v_3 = (-1, 1)$
$c_1 = (0, 0)$
$c_2 = (0, 1)$
$c_3 = (1, 0)$

Assume for this question that e (the mathematical constant) is equal to 3.

(a) What is the distribution over context words for v_1?

(b) What is the distribution over context words for v_2?

(c) Suppose you have two other words 4 and 5. These two words co-occur with each other: 4 occurs with 4, 4 with 5, and 5 with 5. Neither cooccurs with the existing words. What is the minimum number of extra dimensions you need to represent these probabilities while preserving the existing relations? Briefly justify your answer, but you do not need to construct actual vectors.

[2.5 + 2.5 + 2 = 7]